

João Paulo Antunes de Souza

Full Stack Developer · Computer Engineer

Nationality: Brazilian

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Work model: Remote only

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ABOUT ME

Computer Engineer and Full Stack Developer with 6+ years of experience turning field data into useful software for precision agriculture. I build cloud systems that collect, process and display data from farm machines, sensors and maps, and I own each delivery end to end — from database to screen, from first version to production. An electronics background helps me understand the equipment behind the data, and I make technical decisions with cost and business outcome in mind. I use AI tools (Claude Code, ChatGPT) daily, always reviewing and testing their output.

WORK EXPERIENCE

08/2022 – Present

Full Stack Developer (Mid-level)

Zait (formerly Smart Agri / Smart Sensing) — Piracicaba, Brazil

Cloud software for precision agriculture, delivered end to end.

- Built cloud systems that collect, process and visualise data from farm machines, sensors and imagery in real time (Python, AWS, MQTT, React).
- Delivered spatial data solutions: interactive maps, field work planning and ready-to-use QGIS / QField projects (GeoJSON, Shapely, GeoTIFF, Leaflet).
- Automated reporting: maps, figures and charts rendered in the cloud and delivered without manual work.
- Designed REST APIs and integrations with internal and third-party systems, including automatic notifications.
- Implemented authentication, role-based access control and user management (Amazon Cognito).
- Modelled NoSQL data for geographic information (DynamoDB) and managed infrastructure as code (AWS CDK).
- Automated build, test and deployment pipelines, and reduced cloud cost through metric-driven optimisation.

12/2021 – 08/2022

Full Stack Developer (Junior)

Zait (formerly Smart Agri) — Piracicaba, Brazil (hybrid)

- Map-based web interfaces in React and Leaflet, with GeoJSON and drone imagery layers.
- REST APIs in Python (FastAPI) on AWS Lambda and API Gateway with DynamoDB, S3 and Cognito; unit tests; PostgreSQL, MySQL and DynamoDB data modelling.

12/2020 – 12/2021

IT Intern

Zait (formerly Smart Agri) — Piracicaba, Brazil

- REST APIs in Python (FastAPI) and React interfaces; Docker and AWS serverless services (Lambda, API Gateway, DynamoDB, S3).
- Database modelling and unit testing.

06/2019 – 09/2020

Junior Systems Analyst

Farm Solutions — Piracicaba, Brazil

- Web development (HTML, CSS, JavaScript) with Git version control.

EDUCATION AND TRAINING

01/2018 – 12/2023

Bachelor's Degree in Computer Engineering

Federal Institute of Education, Science and Technology of São Paulo (IFSP) — Brazil

- Competitive programming training group.
- Windows application development group (C#, MySQL).

2015 – 2017

Technical Course in Electronics

Etec Prof. Armando Bayeux da Silva — Brazil

LANGUAGE SKILLS

Mother tongue: Portuguese

	Listening	Reading	Spoken production	Spoken interaction	Writing
English	B1	C1	B1	B1	B2

Levels: A1/A2 basic · B1/B2 independent · C1/C2 proficient (Common European Framework of Reference)

DIGITAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C, C#

Front end: React, Next.js, React Native, Leaflet, HTML, CSS

Back end & cloud: FastAPI, REST APIs, AWS (Lambda, API Gateway, DynamoDB, Cognito, S3, IoT Core, CloudWatch), AWS CDK, Docker, MQTT

Data & GIS: NoSQL data modelling, PostgreSQL, MySQL, GeoJSON, Shapely, GeoTIFF, QGIS, QField, Matplotlib

DevOps & quality: Git, GitHub, CI/CD, pytest, Linux

Hardware: Electronics background; MicroPython, ESP32, Arduino, Raspberry Pi

AI tools: AI-assisted development with Claude Code and ChatGPT

PERSONAL PROJECTS

Event promoter app (PWA): Offline-first mobile app to log the workday in seconds: Google sign-in, PDF export, AI form filling from a photo, push reminders and automatic manager spreadsheets. In real use. (TypeScript, serverless, PostgreSQL, Claude API)

Wireless screen casting: Casts a computer screen to phones or browsers, locally or over the internet, with QR-code pairing. (Node.js, WebRTC, WebSocket, Cloudflare Workers)

Running app: Web app for runners with maps and Google sign-in, on its own API. (React, TypeScript, Leaflet, FastAPI, PostgreSQL)

Finance Telegram bot: Logs expenses through a guided chat into a cloud database. (Node.js, serverless, PostgreSQL)

Business website (freelance): Responsive site for a healthcare professional. (Next.js, Tailwind CSS)

Oh My Map: Web map with area drawing, address search and routing (Next.js, Leaflet) — oh-my-map.vercel.app